PRISM-ANT Patent Specification

Simple technical claim brief - not legal advice

Patent specification subject: PRISM-ANT itself - the ordered method that converts model outputs into a safe, deadline-aware, fully utilised station power split.

What should be claimed

Do not claim “dynamic EV power allocation” alone. Published patents already cover dynamic allocation using SOC, temperature, time, and priority. The possible claim focus is the **specific PRISM-ANT order and data flow** below.

The proposed technical invention

PRISM-ANT part	Specific technical feature
Model evidence	Each allocation carries named M1-M6 parameters, confidence and time of calculation.
Safety first	A safety mask produces an inviolable maximum power before scoring or emergency priority.
Readiness reserve	The system calculates and grants the minimum safe kW needed for a vehicle to meet its deadline before sharing extra kW.
Forecast-deadline interlock	Station forecast affects an EV only when that EV has readiness risk; it is not an equal bonus for every vehicle.
Emergency rule	Emergency bypasses normal battery-life preference only. Current and predicted temperature, vehicle, charger, and feeder constraints remain binding.
Event reallocation	Arrival, completion, temperature change, or priority change triggers a new reserve and share calculation.
Explanation record	The user receives assigned kW, safe cap, reserve kW, priority terms, and model reason together.

Simple independent-claim direction

A computer-implemented method comprising: receiving EV telemetry and a charging request; deriving a named predictive signal vector; determining a maximum safe power cap before priority scoring; calculating a deadline readiness reserve; assigning the reserve subject to station capacity; allocating remaining capacity by a priority value; repeating after a station event; and outputting an explanation record identifying the cap, reserve, signals and allocated power.

Comparison with existing approaches

Existing approach	What is already known	PRISM-ANT candidate difference
Dynamic current sharing	Existing patents describe changing allocations using session attributes, SOC, temperature, time and priority.	PRISM-ANT uses a declared safety-mask -> readiness-reserve -> fair-share sequence with evidence carried to the explanation.
Temperature-aware priority	Existing systems can consider temperature when giving priority.	PRISM-ANT treats current and predicted temperature as hard masks, not a score a higher-priority EV can compensate for.
Reallocation on session end	Existing systems can return released current to other sessions.	PRISM-ANT recalculates both the deadline reserve and remainder share from the new model signal vector.

Patent reality check

The comparison identifies a **candidate differentiation**, not proof of patentability. Published examples include US 11,433,772 B2, US 9,469,211 B2, and US 8,643,330 B2. A patent professional must conduct a full prior-art search and prepare a claim chart before any filing.

Evidence to keep

Keep versioned source code, test runs, model input/output schema, screen recordings of explanations, station-event logs, metrics demonstrating deadline success and safety-cap compliance, and dates of development. These are practical materials for counsel to assess the claim.

PRISM-ANT Patent Considerations

PRISM-ANT is the technical subject of the proposed patent specification. The table separates possible claim scope from legal conclusions.

Potential claim element	Why it matters
Ordered control sequence	Safety mask -> readiness reserve -> fair share -> event reallocation is more specific than a generic priority queue.
Named model provenance	Each allocation binds M1-M6 outputs, confidence, timing, cap source, reserve, and final command.
Forecast-deadline interlock	The forecast is tied to individual readiness risk, rather than a station-wide priority uplift.
Emergency semantics	Only battery-life preference is relaxed; physical safety constraints remain non-bypassable.
Explanation coupled to command	The explanation record and allocation command describe the same decision event.
Evidence and validation	Keep model versions, logs, tests, safety results, and a professional prior-art claim chart.

Do not claim alone

SOC-based allocation, temperature awareness, deadlines, emergency priority, or dynamic redistribution individually are not sufficient claim themes because comparable concepts are disclosed in existing EV charging literature and patents.

Claiming boundary

Present PRISM-ANT as a candidate software method. A patent professional must determine novelty, non-obviousness, jurisdiction, claim language, and filing strategy.